Digital Forensics 1 Rubric 5

Forensic Tools & Documentation
The project must include proper use of at least three different forensic tools (EnCase, Autopsy, FTK, SIFT, Volatility) to collect/analyze evidence. Create a forensic case using a case management system. Document findings in a formal report following industry standards. Organize evidence in a structured format suitable for legal review. Include methodology explanations, tool configurations, and chain of custody references.

To satisfy the **Forensic Tools & Documentation** section of your rubric, you must shift your focus from active technical analysis to synthesis, administrative control, and legal defensibility. This rubric evaluates your ability to manage a case using professional workflows and package your findings into a courtroom-ready **Formal Forensic Report**.

The primary objective of this phase is ensuring admissibility. A brilliant technical finding is useless if the tool configurations aren't documented, or if the chain of custody breaks.

🧰 1. Multi-Tool Integration & Configurations

Your project must explicitly highlight the use of **at least three distinct tools** spanning different scopes of the investigation. You must document the exact tool versions, environments, and configuration choices to allow another examiner to precisely replicate your steps.

- **Example 3-Tool Workflow:**
    1. **FTK Imager (v4.7):** Used for physical disk imaging and RAM capture under software write-blocking conditions.
    2. **Autopsy (v4.21):** Used as the primary file system analyzer to parse the Master File Table ($MFT), extract registry hives, examine user folders, and parse web history databases.

3. **Volatility 3 (v2.5):** Executed in a **SIFT Workstation** Linux environment to analyze the `mem.raw` capture, checking for hidden processes (`pslist`/`pstree`) and active network sockets (`netscan`).

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- **Documentation Requirement:** Dedicate an appendix or introductory section to **Tool Configurations**. Do not just state the tool name; list the exact patch versions, specific ingest modules selected (e.g., *Recent Activity, Hash Lookup, Extension Mismatch* in Autopsy), and plugin syntax (e.g., `vol -f mem.raw windows.pslist`).

📂 2. Forensic Case Management Setup

You must document that the investigation was initialized and managed inside a structured environment rather than in a loose folder hierarchy.

- **Using Autopsy/FTK as Case Management:** When initializing a case in Autopsy or FTK, you are prompted for metadata. Take a screenshot of this setup screen.
- **Case Details to Explicitly State:**
    - **Case Number / Unique ID:** e.g., `CASE-2026-098X`
    - **Examiner Name & Agency:** Your Name / Student ID
    - **Case Management Tool Configuration:** Detail how directories are separated. For example, explain how you enforced a strict folder architecture on your storage drive:
        - `\Case_Root\Evidence_Original\` (Write-blocked, read-only copies)
        - `\Case_Root\Autopsy_Project\` (Database and cache)
        - `\Case_Root\Exports\` (Extracted files, carved graphics)
        - `\Case_Root\Documentation\` (Chain of Custody, notes, logs)
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⛓️ 3. Legal Review Organization & Chain of Custody (CoC)

Your evidence must be presented with a clear chain of custody indicating exactly who had access to the data, when they accessed it, and what actions were performed.

- **The Chain of Custody Log:** Include a copy-pasteable or filled-out standard CoC form in your report.

```
                                CHAIN OF CUSTODY LOG
Case Number: CASE-2026-098X                                  Location: Server
Room B, HQ
Item No.  Description            Source System         Serial Number
Hash (MD5)
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Item 01   Physical Disk Image    Workstation Win10-02  SN: 987654XYZ
a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6
Item 02   Volatile Memory Capture Workstation Win10-02  N/A
z9y8x7w6v5u4t3s2r1q0p9o8n7m6l5k4

Item No.   Released By (Name)    Received By (Name)    Date / Time (UTC)
Purpose of Transfer
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Item 01    Officer J. Doe        Your Name             2026-10-29 13:00
Secure transport to Forensic Lab
Item 01    Your Name             Forensic Workstation  2026-10-29 14:15
Ingest into Autopsy Case
```

📝 4. Formal Report Industry-Standard Structure

To get maximum points for this final rubric segment, arrange your actual written document following standard digital forensics reporting structures:

1. **Executive Summary:** A high-level, non-technical overview summarizing the incident, critical findings, and the immediate impact for stakeholders and management.
2. **Evidence Received:** A formal list of physical items or logical images provided, including serial numbers, file sizes, and verification hashes.

3. **Forensic Methodology:** Detailed step-by-step prose explaining your investigative lifecycle, referencing the **Order of Volatility** and **Write-Blocking mechanisms** used to preserve the evidence integrity.
4. **Analysis & Key Findings:** The bulk of my report contains the artifact analysis, tools used, specific registry values parsed, and the **Unified Timeline of Events** mapping out host and network activity.
5. **Conclusion & Recommendations:** Actionable remedial security steps for the organization based on the gaps identified during your investigation.
6. **Appendices:** Tool logs, the full master timeline table, full chain of custody sheets, and software/hardware inventories.